Weekly Hospital Beds occupancy Data

From healthdata.gov

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Introduction

This dataset provides weekly aggregated hospital bed occupancy data from national and state/region levels, reported to the CDC's NHSN. It includes metrics on hospital capacity, occupancy, hospitalizations, and new admissions related to respiratory illnesses like COVID-19, influenza, and RSV. The data tracks healthcare system stress and capacity trends across acute care and critical access hospitals in the U.S., with data collection evolving from mandated to voluntary reporting periods between 2020-2024.

For more details, visit the CDC's NHSN Hospital Respiratory Data page.

Problem Description

This study aims to provide insights into trends, relationships, and predictive patterns.
It focuses on:

- **data visualization,**
- **hypothesis testing,**
- **and regression analysis**

to better understand **hospital capacity and system stress** across different regions and time periods.

Cleaning the data

Week.Ending.Date	Geographic.aggregation	Number.of.Inpatient.Beds.Occupied	Total.COVID.19.Admissions	Total.Influenza.Admissions	Total.RSV.Admissions	Total.Adult.RSV.Admissions
1/6/2024	AK	1014	36	5	3	0
1/6/2024	AR	5286	439	234	3	3
1/6/2024	AZ	11340	645	907	10	8
1/6/2024	CA	48518	3847	1470	38	27
1/6/2024	CO	6955	477	308	3	3
1/6/2024	CT	5916	488	167	13	13
1/6/2024	DC	2306	86	55	0	0
1/6/2024	DE	2009	110	48	4	4
1/6/2024	FL	42465	2143	1414	0	0
1/6/2024	GA	16730	1130	758	1	1
1/6/2024	HI	1903	108	40	2	2
1/6/2024	IA	4681	385	94	9	4
1/6/2024	ID	1964	210	157	2	2
1/6/2024	IL	20043	1602	511	9	9
1/6/2024	IN	10881	979	521	6	5
1/6/2024	KS	4746	389	93	16	7
1/6/2024	KY	9194	521	296	2	2
1/6/2024	LA	8331	417	304	3	3
1/6/2024	MD	8360	612	322	0	0
1/6/2024	ME	2155	150	58	6	6
1/6/2024	MI	16760	1300	497	5	3
1/6/2024	MO	12007	901	345	0	0
1/6/2024	MS	5124	232	263	0	0
1/6/2024	MT	1743	161	129	0	0
1/6/2024	ND	1438	83	52	0	0
1/6/2024	NE	3313	237	50	2	2

- Remove rows with more than 30% missing values
- Keep relevant columns based on business questions

Descriptive statistics

Hospital Occupancy:

- Avg. occupied inpatient beds: 22,287, but highly variable (max: 556,240).
- Significant fluctuations during and after COVID-19.

COVID-19 Admissions:

- Avg. 667 per unit (median 146), peaking at 37,771.

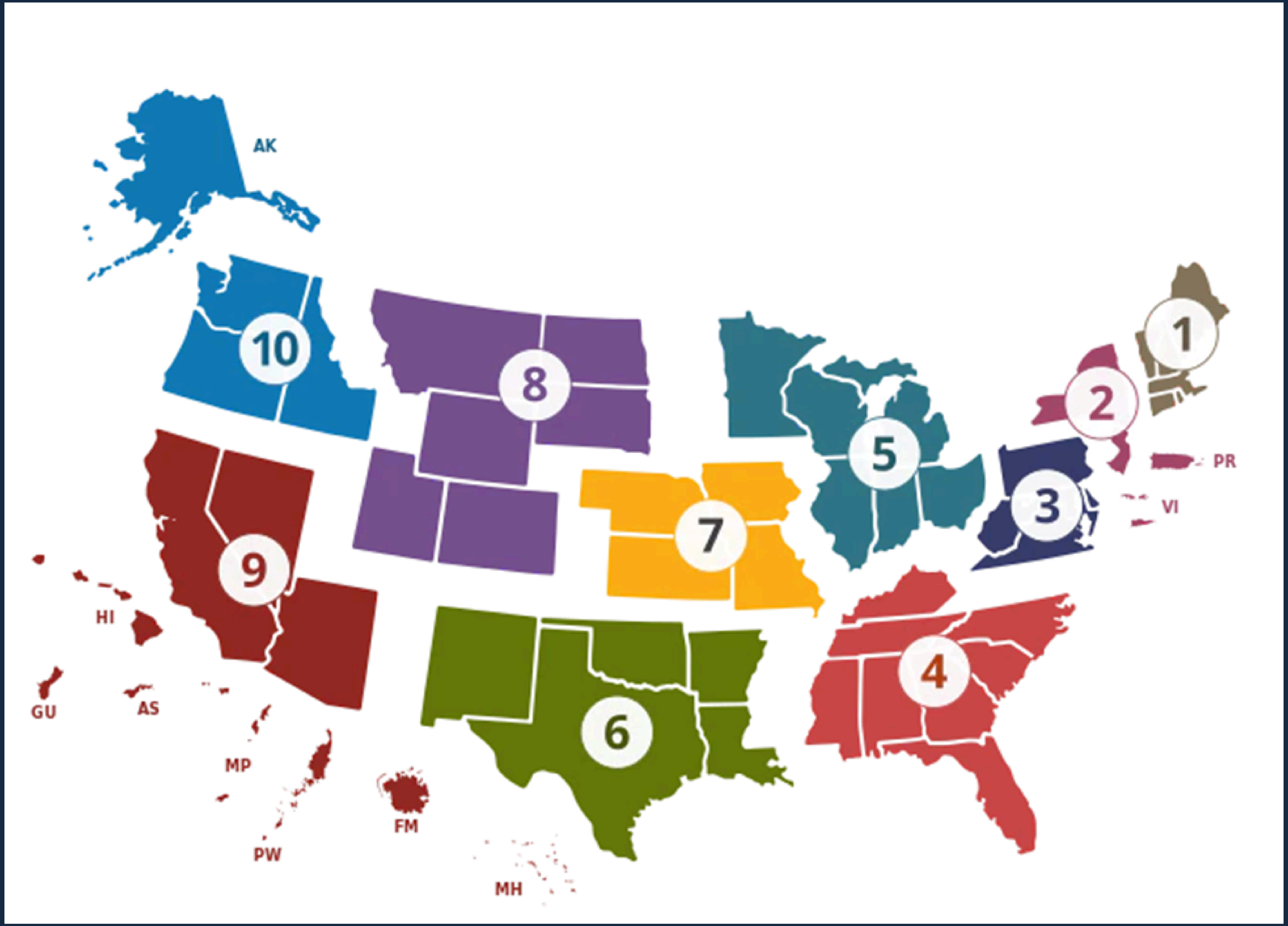
Influenza & RSV Admissions:

- Influenza: Avg. 473 (median 47), max 50,632.
- RSV: Avg. 106, but highly unpredictable (max: 15,314).

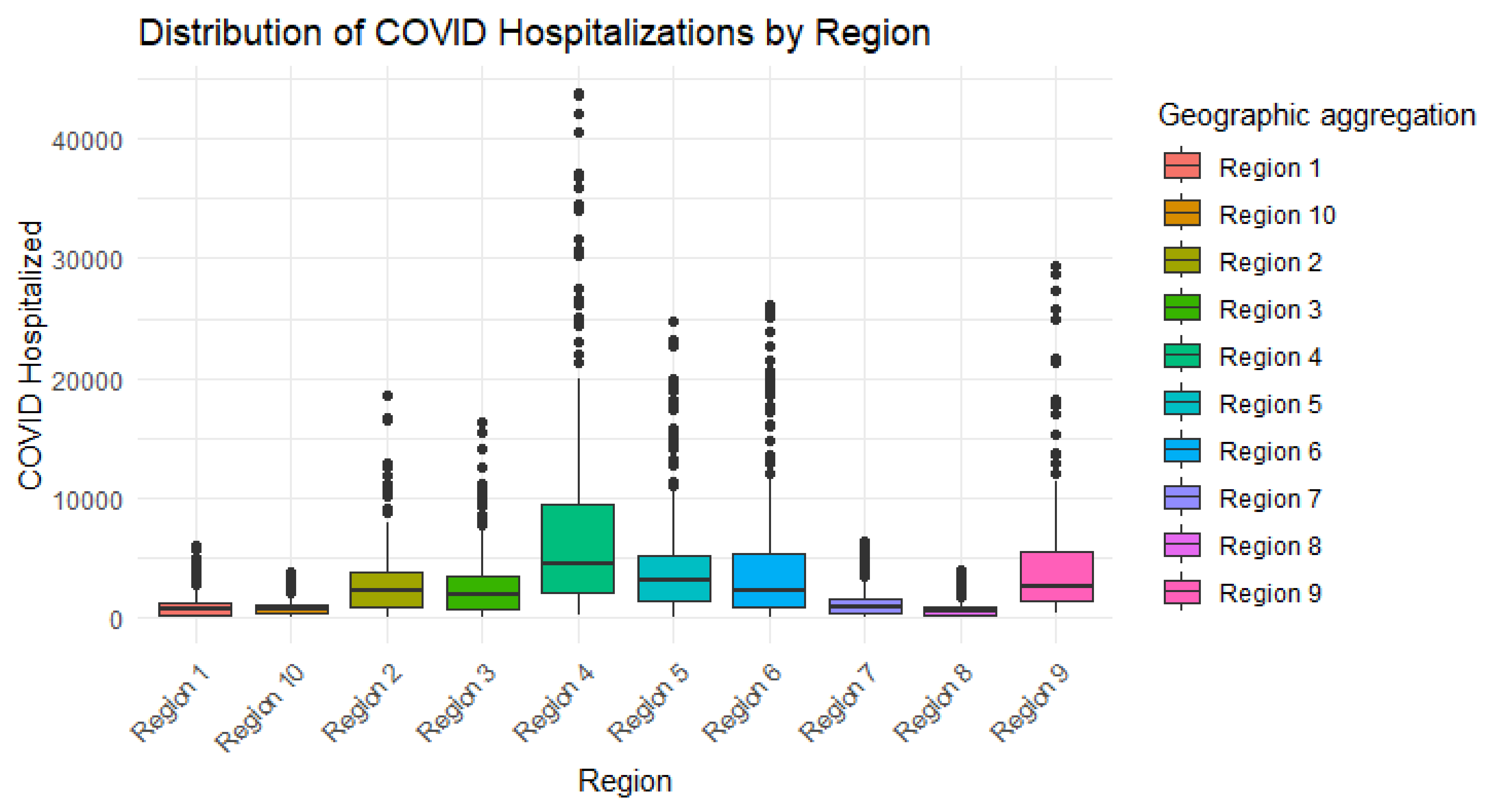
Variability & Trends:

- RSV admissions show highest fluctuations (CV = 5.79), making prediction harder.
- Hospital stress peaks during outbreaks, highlighting the need for trend forecasting & resource planning.

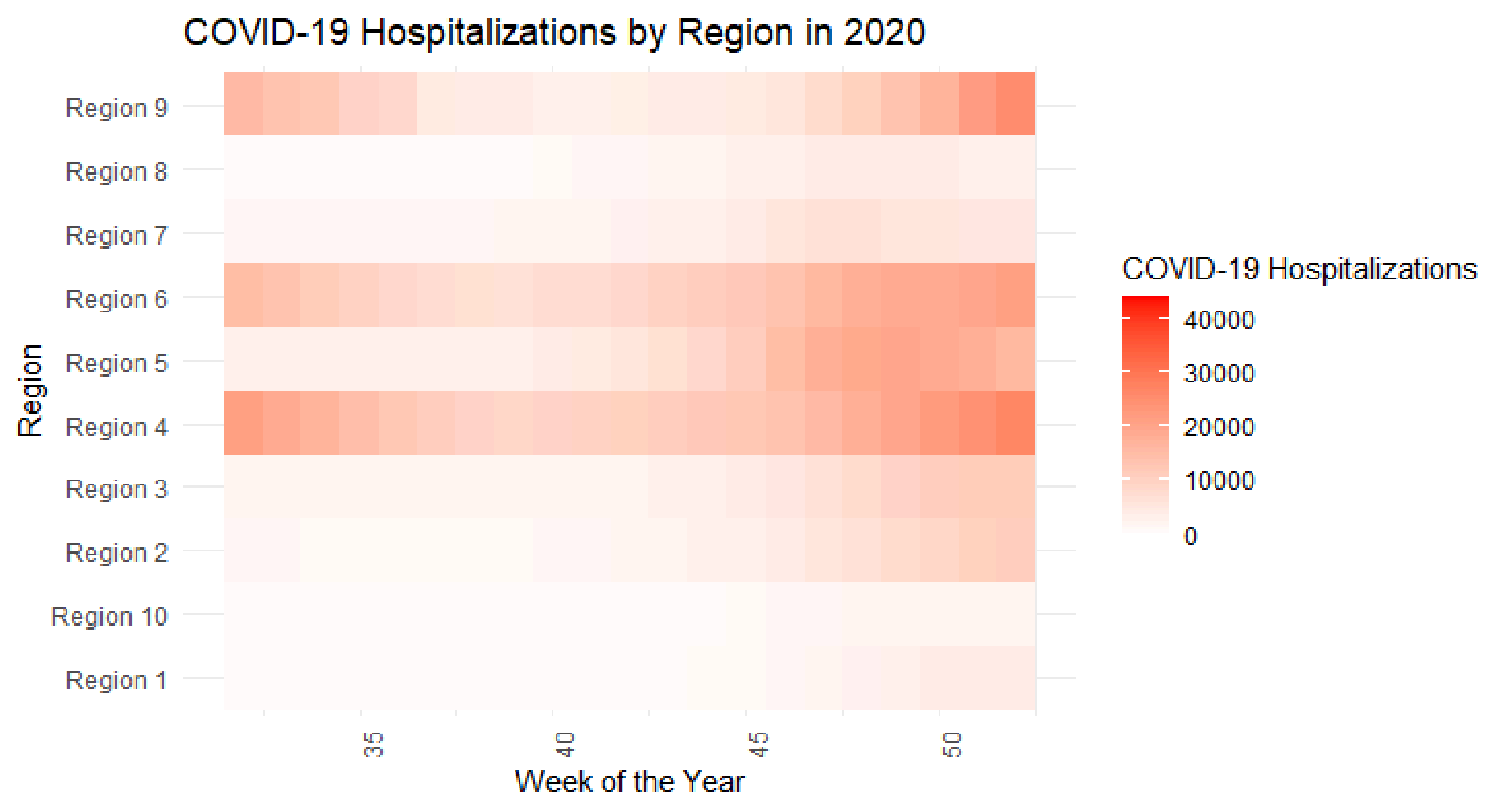
Region Breakdown

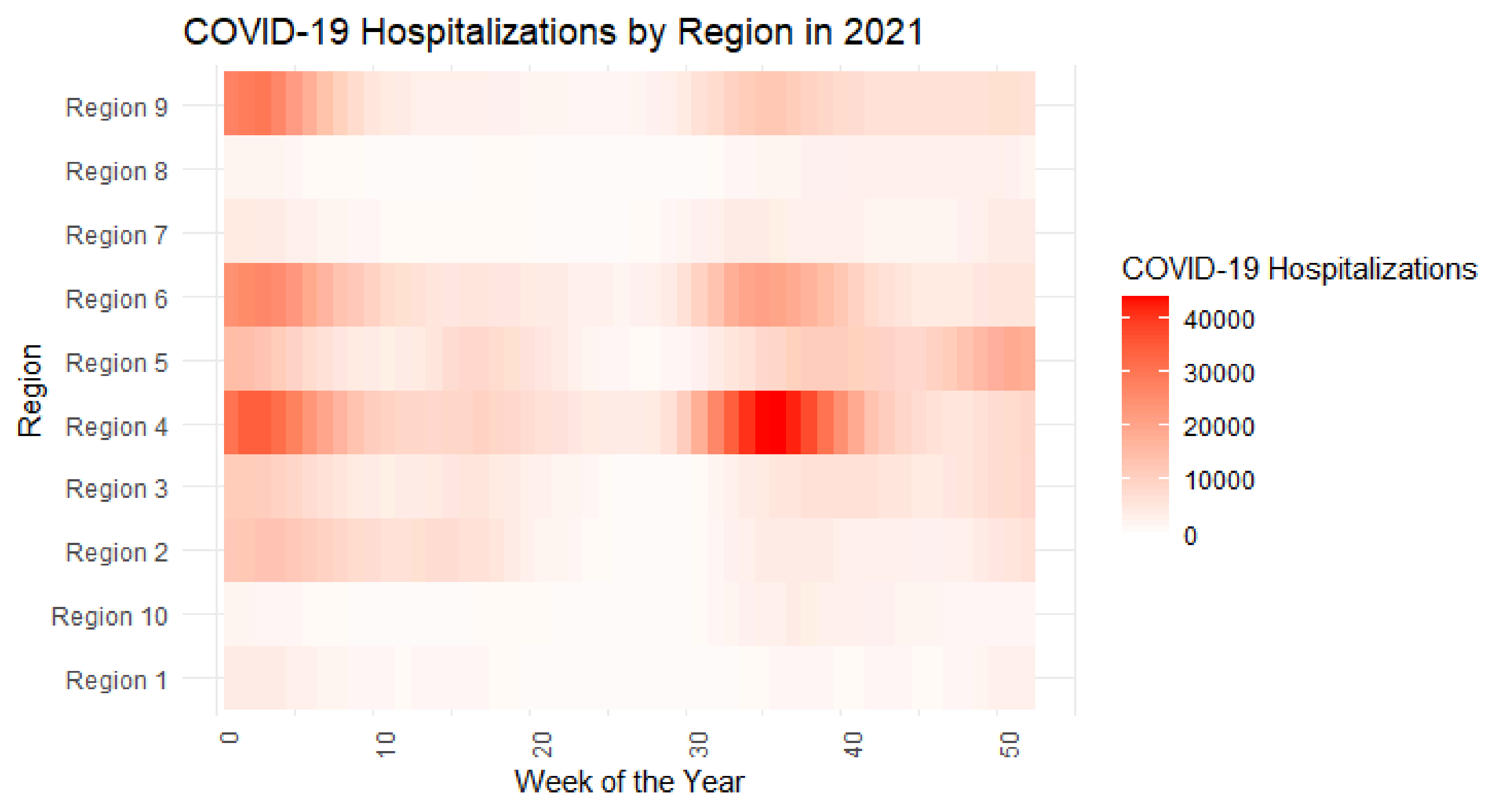


COVID-19 Hospitalizations By Region

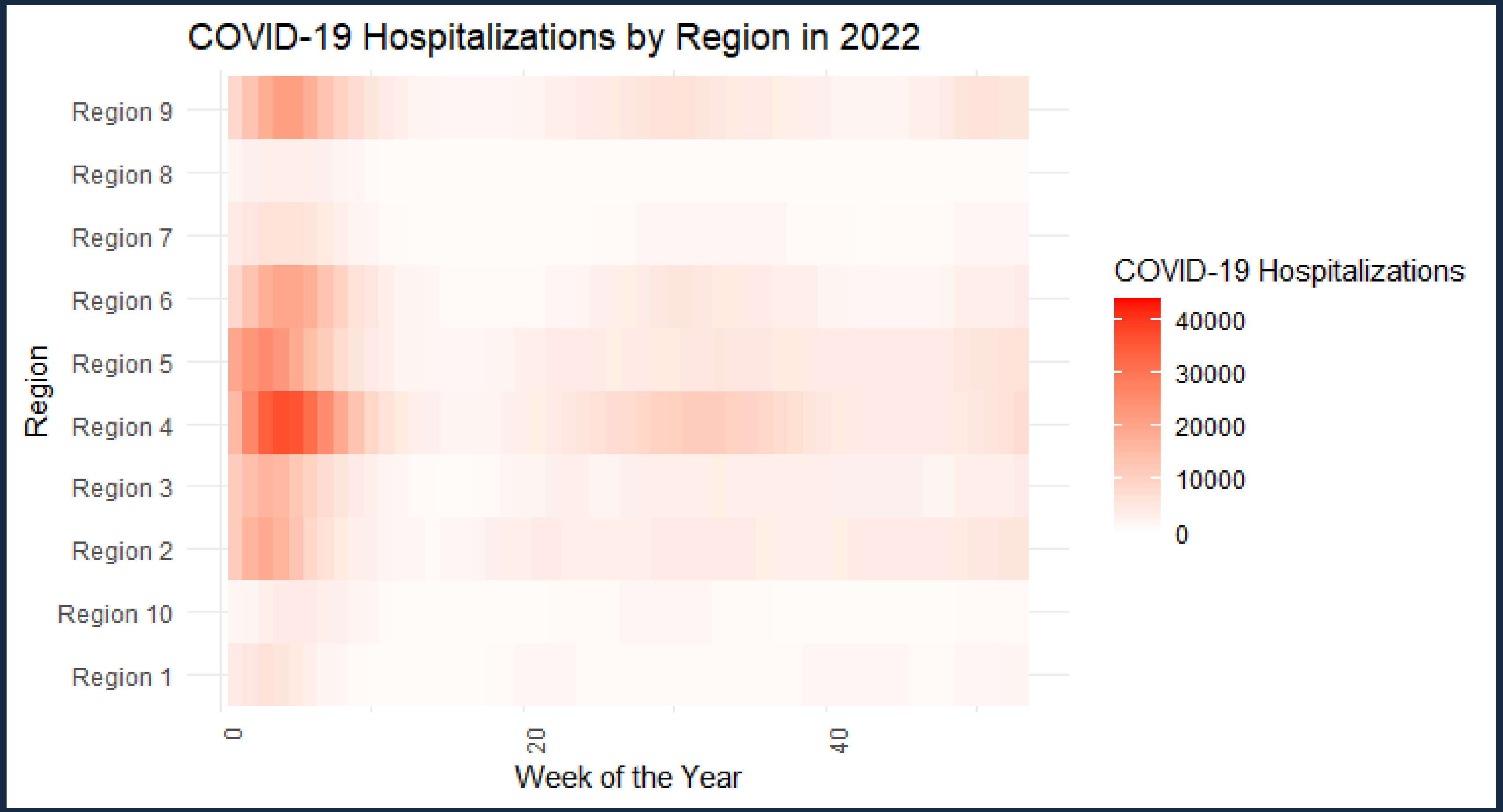


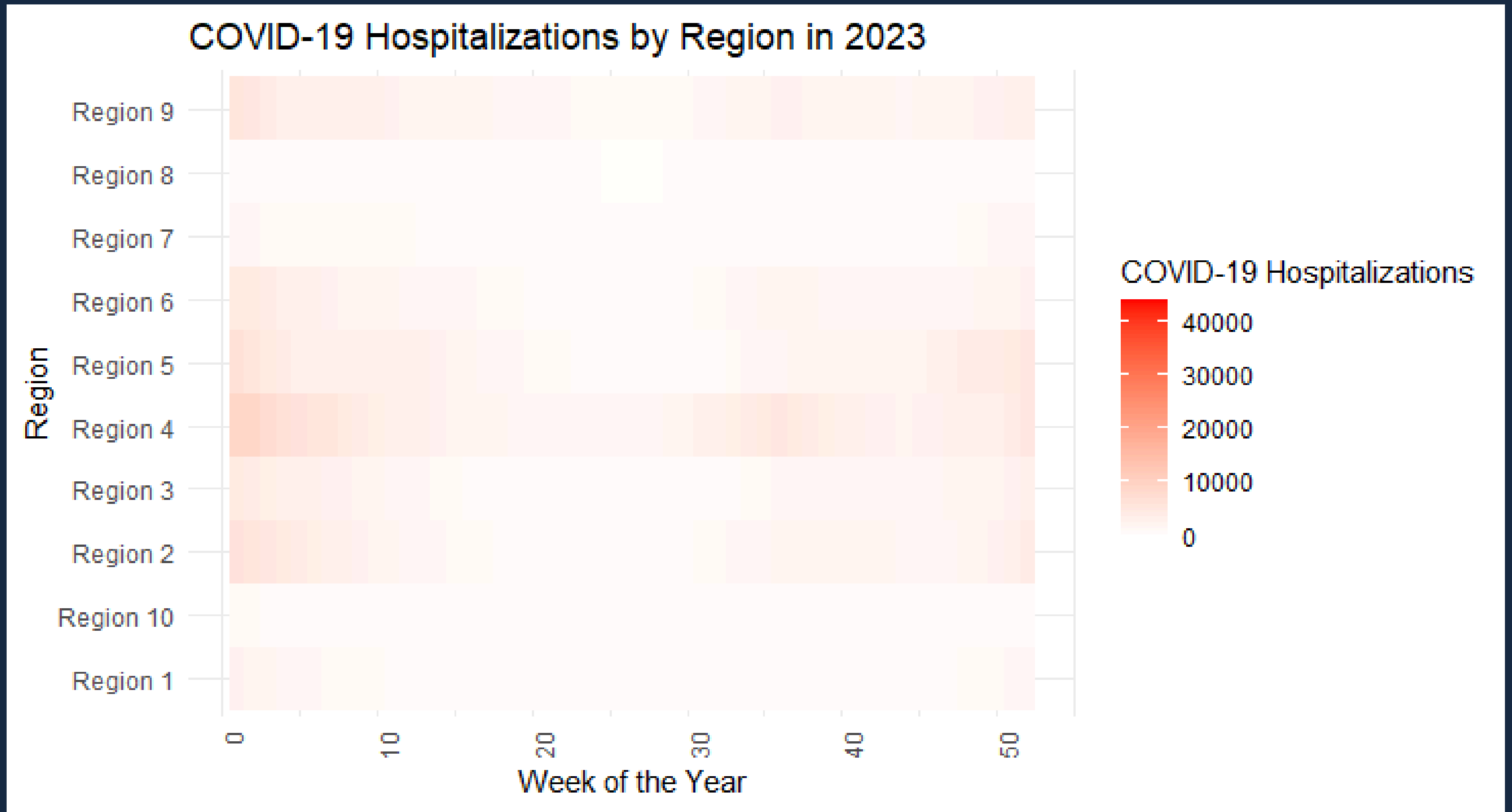
COVID-19 Hospitalization Heat Maps



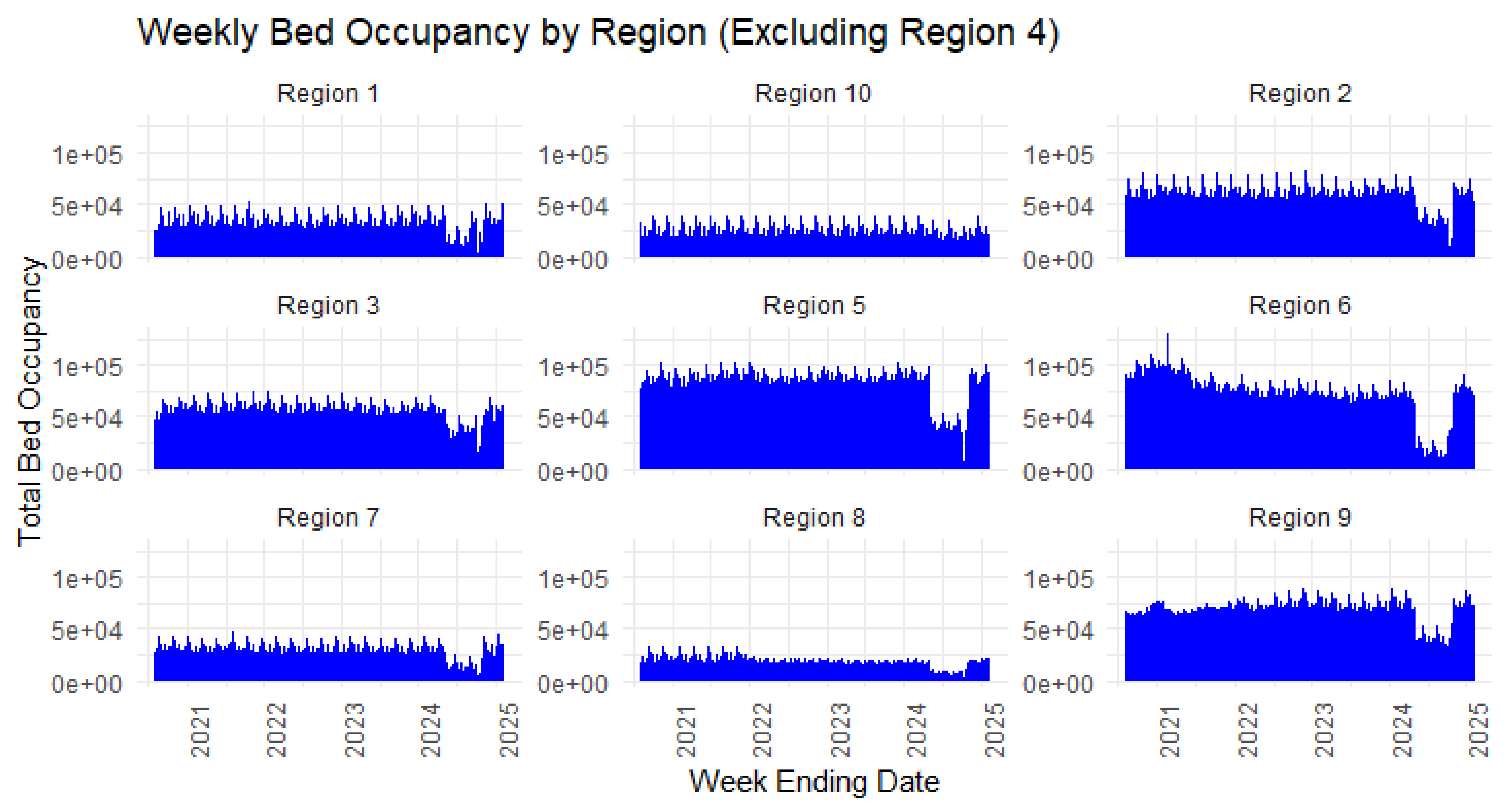


COVID-19 Hospitalization Heat Maps





Weekly Bed Occupancy by Region



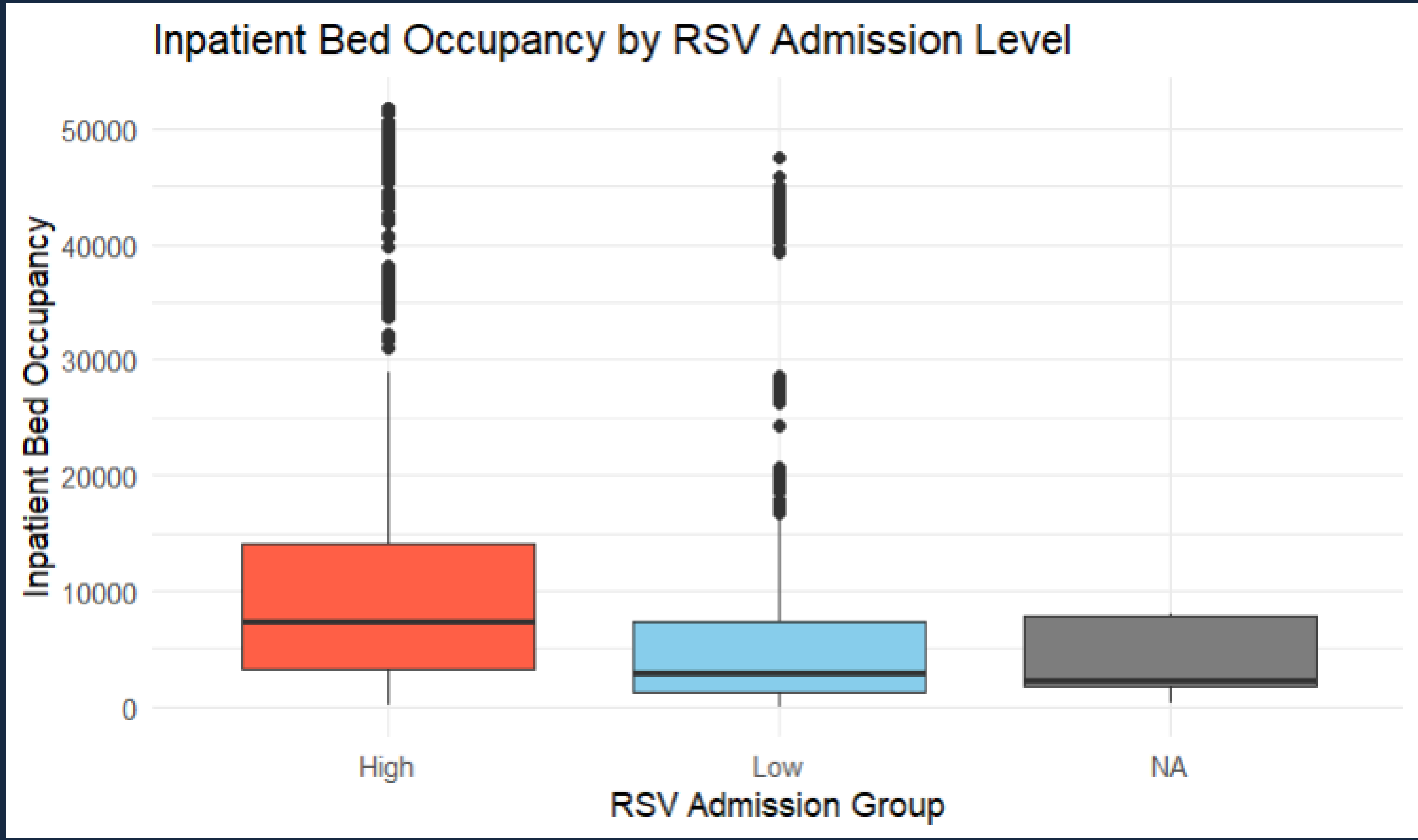
Hypothesis Testing

Do hospitals with higher RSV admissions have higher inpatient bed occupancy?

Test Used: Independent t-test

- Results: $t = 13.08$, $p < 2.2e-16$ (significant)

Conclusion: Hospitals with higher RSV admissions have significantly higher bed occupancy.



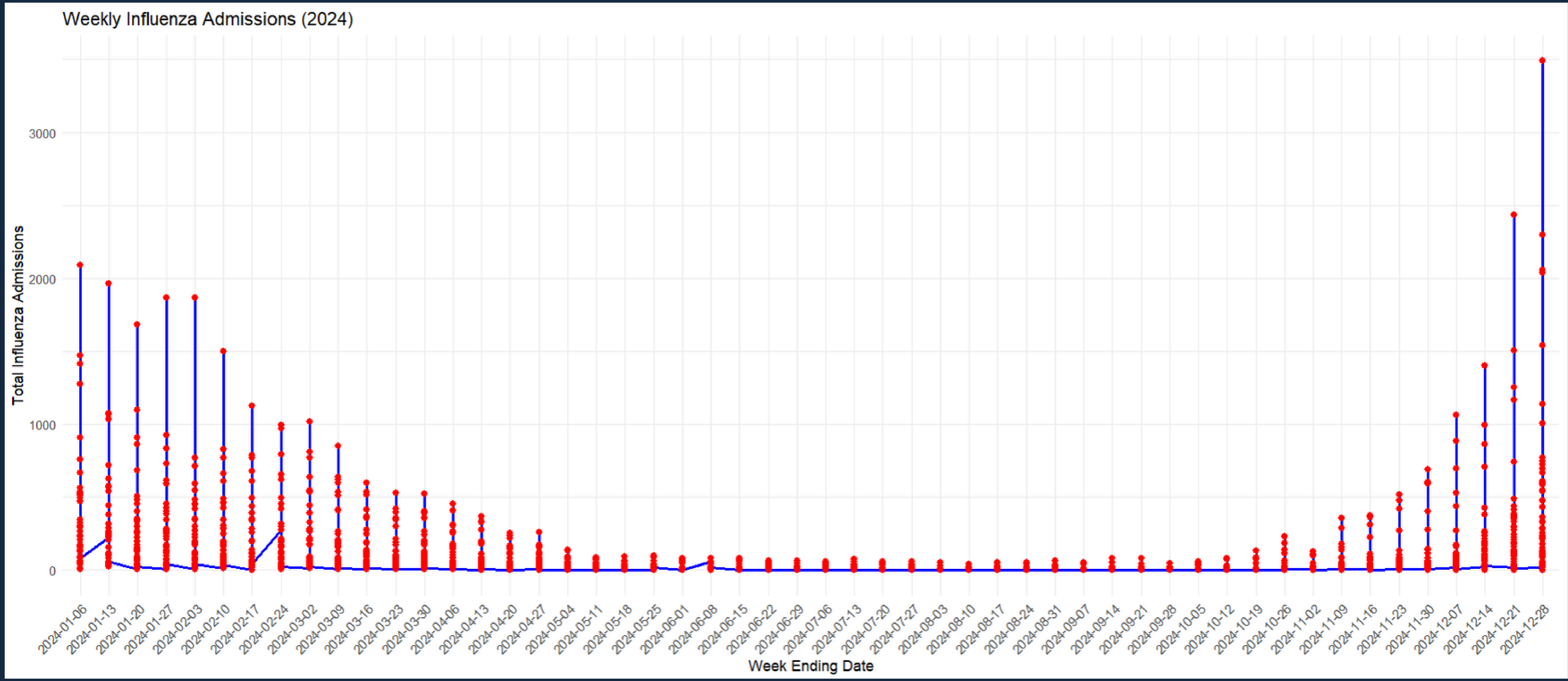
Hypothesis Testing

Do influenza admissions vary significantly between weeks?

Test Used: ANOVA

- Results:F = 14.66, p < 2e-16 (significant)

Conclusion: Influenza admissions fluctuate significantly between weeks, likely due to seasonal trends.



Regression Analysis

- Multiple regression analysis model to examine relationship between hospital bed occupancy and the following dependent variables:
  - COVID-19 Hospitalizations
  - RSV Hospitalizations
  - Influenza Hospitalizations
- Our model excluding outliers is able to predict over 88% of the variability in hospital bed occupancy
  - If outliers were kept in the model the R-squared value dropped to .0188
  - No multicollinearity was found

Predictor	Estimate	Std. Error	t-value	p-value	Interpretation
Intercept	240,900	11,680	20.62	< 2e-16	Baseline when all predictors are zero. Highly significant
COVID_Hospitalized	0.961	0.061	15.68	< 2e-16	Each additional COVID patient increases occupancy by ~0.96 beds. Highly significant
Flu_Hospitalized	6.344	0.539	11.76	< 2e-16	Each additional Flu patient increases occupancy by ~6.34 beds
RSV_Hospitalized	4.126	2.002	2.061	0.0394	Each RSV patient increases occupancy by ~4.13 beds (statistically significant but weak effect)
Region Effects	Varies	Varies	Varies	Significant for most	Regions 4, 5, and 9 have significantly higher occupancy compared to others
Week	-11.29	0.6	-18.73	< 2e-16	Occupancy decreases by approximately 11 beds per week, possibly due to seasonal trends. Overall occupancy shows a decreasing effect over time

Key Findings

- COVID-19, Influenza, and RSV hospitalizations significantly impact inpatient bed occupancy.
    - Hospitals are more burdened by flu and RSV patients than by COVID patients.

- Significant Regional Differences in Bed Occupancy
    - Resource allocation should be region-specific, focusing on areas with the highest burden.

- Seasonal Trend – Occupancy is Declining Over Time
    - The model is statistically robust, meaning these factors (hospitalizations, region, time) accurately predict occupancy.

Conclusion and Future work

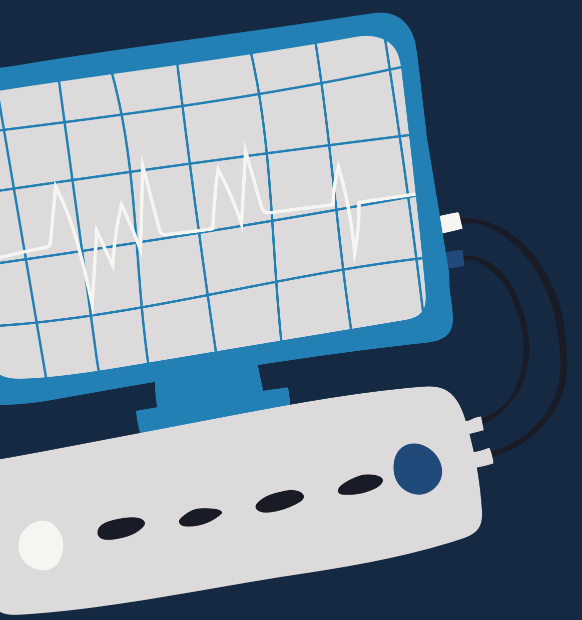
An ongoing journey

- Future work can include examining additional illnesses to help explain seasonal trends
- Exploring healthcare staffing levels against historical trends
  - Improved staffing can help alleviate stress for healthcare workers

- In conclusion, the data we examined identified trends that can be used by many stakeholders including:
  - Healthcare officials
  - Federal and Local Governments
  - Healthcare Planning Officials

Questions or comments?

References



Centers for Disease Control and Prevention. (2025). Weekly Hospital Beds Occupancy - Regional Level [Dataset]. Retrieved February, 2025, from [https://www.cdc.gov/].

This presentation includes AI-generated content from ChatGPT, an OpenAI language model.